

SAFETY DATA SHEET

Section 1: IDENTIFICATION

Product Name: DEMAND
Design Code: A12689G
Recommended Use: Insecticide
Company Details: Syngenta Crop Protection Limited
Address: Level 4,
 60 Parnell Road,
 Parnell
AUCKLAND 1052
NEW ZEALAND

Telephone number: (weekdays) 09 306 1500
Emergency Telephone number: (24 Hours) 0800 734 607
National Poisons & Hazchem Information Centre : 0800 POISON (0800 764 766)

Section 2: HAZARDS IDENTIFICATION

Hazard classification: 6.1D, 6.3A, 6.4A, 6.5B, 6.9B, 9.1A, 9.3C, 9.4A
Priority Identifier: WARNING
 KEEP OUT OF REACH OF CHILDREN

Secondary Identifiers:

- 6.1D = May be harmful if swallowed, inhaled or absorbed through the skin.
- 6.3A = May cause skin irritation
- 6.4A = May cause eye irritation
- 6.5B = May cause sensitisation from prolonged skin contact.
- 6.9B = May cause neurotoxic damage from repeated oral exposure at high doses.
- 9.1A = Very toxic to aquatic organisms.
- 9.3C = Harmful to terrestrial vertebrates.
- 9.4A = Very toxic to terrestrial invertebrates.

Section 3: COMPOSITION / INFORMATION ON INGREDIENTS

Mixture:		
Chemical Identity of ingredients:		
Ingredient	CAS no.	Content (% w/v)
Lambda-cyhalothrin	91465-08-6	2.5
Solvent naphtha (petroleum), light aromatic	64742-94-5	>= 1 - < 2.5
phosphoric acid	7664-38-2	>= 1 - < 3
m-tolylidene diisocyanate	26471-62-5	>= 0.25 - < 1
Isocyanic acid, polymethylenepolyphenylene ester	9016-87-9	>= 0.1 - < 1
4,4'-methylenediphenyl diisocyanate	101-68-8	>= 0.1 - < 1
1,2-benzisothiazol-3 (2H)-one	2634-33-5	>= 0.025 - < 0.05
other ingredients determined not to be hazardous	-	to 100%

Section 4: FIRST AID MEASURES

Description of First Aid measures:

General Advice: For advice contact the National Poisons Centre on 0800 POISON (0800 764 766) or a doctor immediately. Begin artificial respiration if the victim is not breathing. Use mouth to nose rather than mouth to mouth. Obtain medical attention.

If inhaled:	Move the victim to fresh air. If breathing is irregular or stopped, administer artificial respiration. Keep patient warm and at rest. Call a Doctor or the Poisons Information Centre immediately.
In case of skin contact:	Take off all contaminated clothing immediately. Wash off immediately with plenty of water. If skin irritation persists, call a doctor. Wash contaminated clothing before re-use.
In case of eye contact:	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses (if present). Immediate medical attention is required.
If swallowed:	If swallowed seek medical advice immediately and show the container or label. DO NOT induce vomiting.
Important symptoms and effects, both acute and delayed:	
Symptoms:	Aspiration may cause pulmonary oedema and pneumonitis. Skin contact paresthesia effects (itching, tingling, burning or numbness) are transient, lasting up to 24 hours.
Indication of any immediate medical attention and special treatment needed:	
	No specific antidote is available. If poisoning is suspected apply symptomatic therapy. Do not induce vomiting: contains petroleum distillates and/or aromatic solvents. Persons suffering a temporary allergic reaction may respond to treatment with anti-histamines or steroid creams and/or systemic steroids.

Section 5: FIRE-FIGHTING MEASURES

Extinguishing media:	
Suitable extinguishing media:	Small fires: Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide. Large Fires: Alcohol resistant foam or water spray.
Unsuitable extinguishing media:	Do not use a solid water stream as it may scatter and spread fire.
Special hazards arising from the substance or mixture:	
Specific hazards during fire-fighting:	As the product contains combustible organic components, fire will produce dense black smoke containing hazardous products of combustion (see section 10) Exposure to decomposition products may be a hazard to health.
Advice for firefighters:	
Special protective equipment for firefighters:	Wear full protective clothing and self-contained breathing apparatus.
Hazchem Code:	2X
Further information:	Do not allow run-off from fire fighting to enter drains or water courses. Cool closed containers exposed to fire with water spray.

Section 6: ACCIDENTIAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures

Refer to protective measures listed in Sections 7 and 8.

Environmental Precautions:

Do not flush into surface water or sanitary sewer system.
If the product contaminates rivers and lakes or drains inform respective authorities.

Methods and material for containment and cleaning up:

Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).
Clean contaminated surface thoroughly.
Clean with detergents. Avoid solvents.
Retain and dispose of contaminated wash water.

Reference to other sections:

Refer to disposal considerations listed in Section 13.
Refer to protective measures listed in sections 7 and 8.

Section 7: HANDLING AND STORAGE

Precautions for Safe handling:

Advice on safe handling:

No special protective measures against fire required.
Avoid contact with skin and eyes.
When using do not eat, drink or smoke.
For personal protection see section 8.

Conditions for safe storage, including any incompatibilities:

Requirements for storage area and containers:

No special storage conditions required. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep out of the reach of children. Keep away from food, drink and animal feedingstuffs.

Specific end use(s)

Specific use(s)

For proper and safe use of this product, please refer to the approval conditions laid down on the product label.

Section 8: EXPOSURE CONTROL / PERSONAL PROTECTION

Control Parameters

Occupational Exposure Limits:

Components	CAS No	Value type (form of exposure)	Control parameters	Basis
lambda-cyhalothrin	91465-08-6	TWA	0.04 mg/m ³ (Skin)	Syngenta
Solvent naphtha (petroleum), light arom.; Low boiling point naphtha - unspecified	64742-95-6	TWA	19 ppm 100 mg/m ³	Supplier
phosphoric acid	7664-38-2	TWA	1 mg/m ³	WES
4,4'-methylene diphenyl diisocyanate	101-68-8	TWA	0.02 mg/m ³ (as -NCO)	WES
		STEL	0.07 mg/m ³ (as -NCO)	WES

Exposure controls	
Engineering measures:	Containment and/or segregation is the most reliable technical protection measure if exposure cannot be eliminated. The extent of these protection measures depends on the actual risks in use. Maintain air concentrations below occupational exposure standards. Where necessary, seek additional occupational hygiene advice.
Personal Protective Protection:	
Eye protection:	No special protective equipment required.
Hand protection:	
Material:	Chemical resistant, such as nitrile rubber.
Break through time:	>480 min
Glove thickness:	0.5 mm
Remarks:	Wear protective gloves. The choice of an appropriate glove does not only depend on its material but also on other quality features and is different from one producer to the other. Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion, and the contact time. The break through time depends amongst other things on the material, the thickness and the type of glove and therefore has to be measured for each case. Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.
Skin and body protection:	Choose body protection in relation to its type, to the concentration and amount of dangerous substances, and to the specific work-place. Remove and wash contaminated clothing before re-use. Wear as appropriate: Impervious clothing.
Respiratory protection:	When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Suitable respiratory equipment: Respirator with combination filter for vapour/particulate (EN 141) The filter class for the respirator must be suitable for the maximum expected contaminant concentration (gas/vapour/aerosol/particulates) that may arise when handling the product. If this concentration is exceeded, self-contained breathing apparatus must be used.
Filter type:	Combined particulates and organic vapour type (A-P)
Protective measures:	The use of technical measures should always have priority over the use of personal protective equipment. When selecting personal protective equipment, seek appropriate professional advice. Personal protective equipment should be certified to appropriate standards.

Section 9: PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties:

Appearance:	Liquid
Colour:	Light beige to brown
Odour:	Mildly phenolic
Odour threshold:	No data
pH value	4 – 8, concentration: 1 % w/v
Melting point / freezing point:	No data
Initial boiling point and boiling range:	No data

Flash point:	> 99°C (1020 hPa) Method: Pensky-Martens closed cup
Flammability:	Not classified as a flammability hazard
Upper flammability / explosive limits:	No data
Lower flammability / explosive limits:	No data
Vapour pressure:	No data
Vapour Density:	No data
Density:	1.03 g/cm ³ at 25°C
Solubility in other solvents:	Soluble in water
Partition co-efficient: n-octanol / water:	No data
Autoignition temperature:	600°C
Decomposition temperature:	No data
Dynamic viscosity:	33 - 263 mPa.s (40°C) 41 - 303 mPa.s (20°C)
Explosive properties:	Not explosive
Oxidising properties:	The substance or mixture is not classified as oxidizing
Surface tension:	49.8 mN/m, 20°C 61.1 mN/m, 20°C 52.7 mN/m, 20°C
Minimum ignition energy:	No data

Section 10: STABILITY AND REACTIVITY

Reactivity:	See Section: "Possibility of Hazardous Reactions".
Chemical Stability:	The product is stable when used in normal conditions.
Possibility of Hazardous Reactions:	Hazardous reactions: No dangerous reaction known under conditions of normal use.
Conditions to Avoid	No decomposition if used as directed.
Incompatible Materials:	Materials to avoid: None known
Hazardous Decomposition Products:	Combustion or thermal decomposition will evolve toxic and irritant vapours.

Section 11: TOXICOLOGICAL INFORMATION

HSNO Classifications:	
6.1D =	May be harmful if swallowed, inhaled or absorbed through the skin.
6.3A =	May cause skin irritation
6.4A =	May cause eye irritation
6.5B =	May cause sensitisation from prolonged skin contact.
6.9B =	May cause neurotoxic damage from repeated oral exposure at high doses.

Acute toxicity (Similar product or calculation)	
Swallowed:	LD ₅₀ Acute toxicity estimate: > 2,000 mg/kg Method: Calculation method.
Dermal absorption:	LD ₅₀ Acute toxicity estimate: > 2,000 mg/kg Method: Calculation method
Inhaled:	LC ₅₀ (4 h) Acute toxicity estimate: 2.44 mg/L Exposure time: 4 h Test atmosphere: dust/mist Method: Calculation method.
Aspiration hazard:	Component:

Respiratory irritation:	Solvent naphtha (petroleum), light arom.; Low boiling point naphtha -unspecified: May be fatal if swallowed and enters airways.
Skin corrosion / irritation:	Not classified
Eye damage / irritation:	IRRITANT (rabbit)
Respiratory or Skin Sensitisation:	IRRITANT (HSNO Classification) (rabbit)
	SKIN SENSITISER (guinea pigs)
Chronic / Long Term Effects (active ingredient)	
Germ cell mutagenicity:	Animal testing did not show any mutagenic effects.
Carcinogenicity:	No evidence of carcinogenicity in animal studies.
Reproductive toxicity:	No toxicity to reproduction.
Specific Organ toxicity:	<i>Single exposure:</i> Target Organs: Respiratory tract The substance or mixture is classified as specific target organ toxicant, single exposure, GHS category 3 with respiratory tract irritation. <i>Repeated exposure:</i> Target Organs: Nervous system The substance or mixture is classified as specific target organ toxicant, repeated exposure, Class 6.9B, (GHS: category 2).
Narcotic Effects:	Component: Solvent naphtha (petroleum), light arom.; Low boiling point naphtha -unspecified: Exposure in poorly ventilated spaces may have narcotic effects.

Section 12: ECOLOGICAL INFORMATION

HSNO Classifications:

- 9.1A = Very toxic to aquatic organisms.
9.3C = Harmful to terrestrial vertebrates.
9.4A = Very toxic to terrestrial invertebrates.

Ecotoxicity Effects - Aquatic

Acute toxicity to fish:	LC ₅₀ (96h) = 0.21 µg/L (<i>Leuciscus idus</i> (Golden orfe)) (active ingredient) LC ₅₀ (96h) = 0.078 µg/L (<i>Lepomis macrochirus</i> (Bluegill sunfish)) (active ingredient)
Toxicity to daphnia and other aquatic invertebrates:	EC ₅₀ (48h) = 0.36 µg/L (<i>Daphnia magna</i> (water flea)) (active ingredient)
Toxicity to algae:	E _r C ₅₀ (96 h) = >1 mg/L (<i>Pseudokirchneriella subcapitata</i> (green algae)) (active ingredient)

Ecotoxicity Effects - Terrestrial

Toxicity to Birds:	LD ₅₀ = >3950 mg/kg (mallard ducks) (active ingredient)
Toxicity to soil dwelling organisms:	LC ₅₀ (14 days) = >1000 mg/kg (earthworms) (active ingredient)
Toxicity to Bees:	LD ₅₀ (48 h, oral) = 38 ng/bee (active ingredient) LD ₅₀ (48 h, contact) = 909 ng/bee (active ingredient)

Persistence and degradability:

Biodegradability:	Not readily biodegradable.
Stability in water:	Degradation half-life (DT ₅₀): 7d Not persistent in water.

Bioaccumulative potential:	
Bioaccumulation:	Lambda-cyhalothrin bioaccumulates.
Mobility in soil:	
Distribution among environmental compartments:	Immobile
Stability in soil:	DT ₅₀ : 56 d Percentage dissipation: 50% Not persistent in soil.
Other adverse effects:	
Results of PBT and vPvB assessment (product):	This substance/mixture contains no components considered to be persistent, bioaccumulating and toxic (PBT). This substance is not considered to be very persistent and very bioaccumulating (vPvB) at levels of 0.1% or higher.

Section 13: DISPOSAL CONSIDERATIONS

Product Disposal:	DO NOT contaminate ponds, waterways or ditches with chemical or used containers. DO NOT dispose of waste into sewer. Dispose of this product only by using according to the label. Otherwise, dispose of waste at an approved landfill or other approved facility that will ensure the substance does not exceed the tolerable exposure limit (TEL) or environmental exposure limit (EEL), where relevant, or will treat the substance so that it is rendered no longer hazardous.
Container Disposal:	Ensure the container is empty. Triple rinse empty container and add rinsate to the spray tank. Recycle empty container through Agrecovery (0800 247 326, www.agrecovery.co.nz). Otherwise crush and bury in a suitable landfill. DO NOT reuse this container for any other purpose.

Section 14: TRANSPORT INFORMATION

Rail / Road (NZS 5433)	UN-No: 3082 Class: 9 Packaging Group: III Proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Lambda-cyhalothrin)
Sea (IMDG-Code)	UN-No: 3082 Class: 9 Packaging Group: III Proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Lambda-cyhalothrin) EmS Code: F-A, S-F MARINE POLLUTANT: Yes
Air (IATA)	UN-No: 3082 Class: 9 Packaging Group: III Proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (Lambda-cyhalothrin) Packing instruction: 964 (cargo and passenger aircraft) Packing instruction (LQ): Y964 (cargo and passenger aircraft)

Section 15: REGULATORY INFORMATION

HSNO Approval Number:	HSR000337
Tolerable Exposure Limit or Environmental Exposure Limit:	No TEL or EEL values are set for this substance at this time
Required Regulatory Controls:	
Certified handler:	No
Tracking:	No
Record Keeping:	Yes, 9.1A substance
ACVM Registration:	Not applicable
ACVM Controls:	Not applicable
International Agreements related to the substance (eg, Montreal Protocol, Stockholm Convention or Rotterdam Convention):	Not applicable

Section 16: OTHER INFORMATION

Date of SDS Preparation / Review:	24 February 2023
Version number of SDS:	6.0
Key / Legend to abbreviations and acronyms used: AICS - Australian Inventory of Chemical Substances; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; CPR - Controlled Products Regulations; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; N.O.S. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WES - Workplace Exposure Standard (Worksafe NZ); WHMIS - Workplace Hazardous Materials Information System	
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